

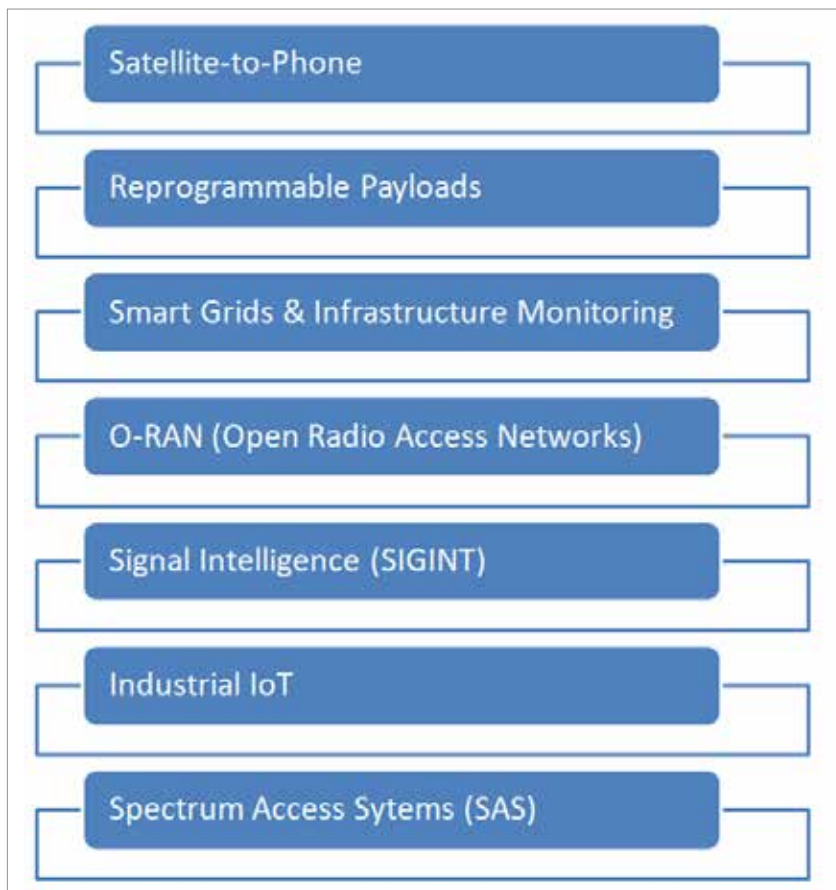


Figure 1: Industrial deployments and use cases of radio and software defined radio (SDR)

Secure identity and authentication with radio fingerprinting: Spoofing is a major risk in software defined radio (SDR), telecom units, military deployments and cognitive radio environments. Radio fingerprinting can be used to associate radio transmitters with a unique hardware fingerprint. This helps to avoid and push back unauthorised phantom stations in secured networks.

6G technology and automated resource trading: As the world moves towards 6G deployments, millions of edge devices and micro-cells will coexist leading to issues of congestion, clashing, crushing of signals, and so on. By using blockchain-based smart contracts, a drone or device can autonomously fetch a few minutes of high-bandwidth radio frequency (RF) spectrum (deployed on a ground station) using a blockchain integrated digital wallet, executing the payment with the instant frequency handoff.

Implementation of DeWi and decentralised antenna design

Rather than millions of towers under one company, in blockchain-based

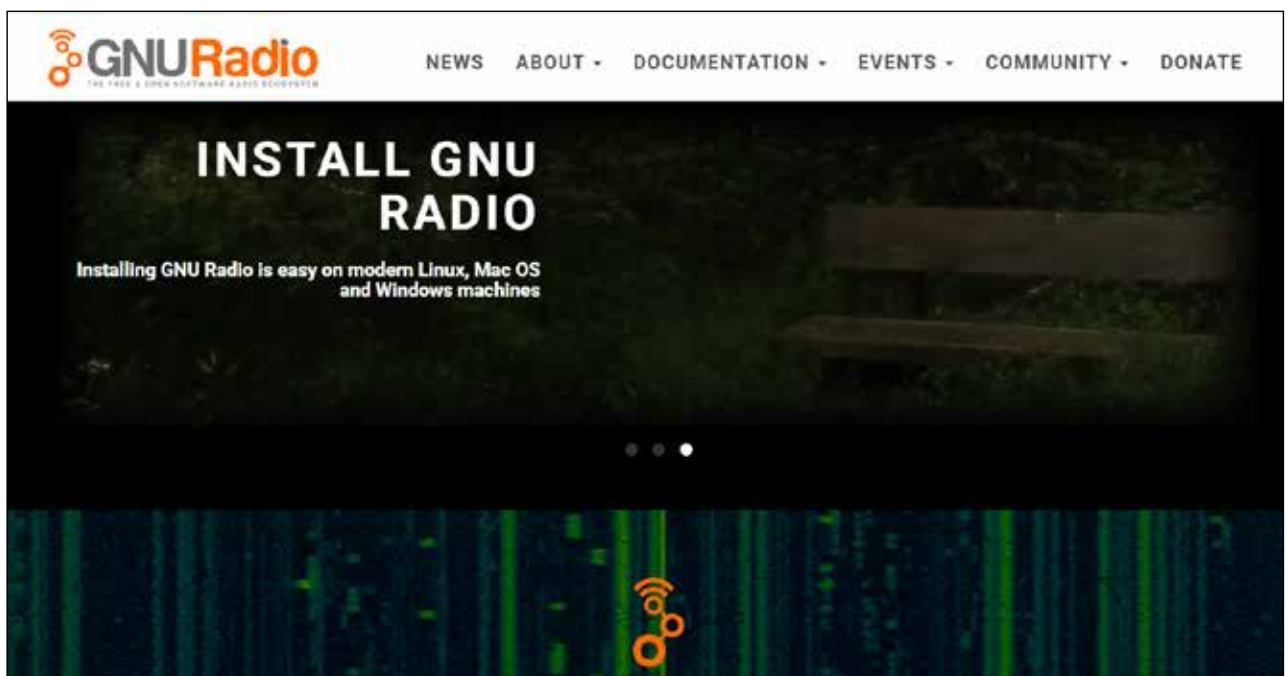


Figure 2: GNU Radio for simulation of signal analysis including blockchain-based decentralised scenarios